RETAIL MARKETING CAMPAIGN ANALYTICS

A retail company wants to analyze customer buying behavior and marketing campaign performance using Snowflake sample data.

The company has three teams:

1. **Marketing Analysts**
2. **Finance Team**
3. **ETL Developers**

Each team should use a separate warehouse. The company also wants to improve query performance using:

1. **Warehouse scaling**
2. **Query result cache**
3. **Cluster key on customer sales date**

Use Snowflake sample data from:

SNOWFLAKE_SAMPLE_DATA.TPCDS_SF10TCL.STORE_SALES

SNOWFLAKE_SAMPLE_DATA.TPCDS_SF10TCL.CUSTOMER

SNOWFLAKE_SAMPLE_DATA.TPCDS_SF10TCL.DATE_DIM

SNOWFLAKE_SAMPLE_DATA.TPCDS_SF10TCL.ITEM

Questions

Part 1: Warehouse Setup

Create three warehouses:

1. **MARKETING_WH for marketing analysts**
2. **FINANCE_WH for finance team**
3. **ETL_WH for ETL developers**

Use different warehouse sizes based on workload.

ANS:

Creating MARKETING_WH warehouse of XSMALL size

```sql
CREATE OR REPLACE WAREHOUSE MARKETING_WH
    WAREHOUSE_SIZE = 'XSMALL'
    AUTO_SUSPEND = 60
    AUTO_RESUME = TRUE
    COMMENT = 'Warehouse for Marketing Analysts';
```

Creating FINANCE_WH warehouse of SMALL size

```sql
CREATE OR REPLACE WAREHOUSE FINANCE_WH
    WAREHOUSE_SIZE = 'SMALL'
    AUTO_SUSPEND = 60
    AUTO_RESUME = TRUE
    COMMENT = 'Warehouse for Finance Team';
```

Creating ETL_WH warehouse of MEDIUM size

```sql
CREATE OR REPLACE WAREHOUSE ETL_WH
    WAREHOUSE_SIZE = 'MEDIUM'
    AUTO_SUSPEND = 120
    AUTO_RESUME = TRUE
    COMMENT = 'Warehouse for ETL Developers';
```

It was successfully executed.

Results (just now)

Table | Chart | Pivot

	status
1	Warehouse MARKETING_WH successfully created.

Results (just now)

Table | Chart | Pivot

	status
1	Warehouse FINANCE_WH successfully created.

Results (just now)

Table | Chart | Pivot

	status
1	Warehouse ETL_WH successfully created.

	name	state	type	size	running	queued	is_default	is_current	auto_suspend
1	COMPUTE_WH	SUSPENDED	STANDARD	X-Small	1	0	Y	N	300
2	ETL_WH	SUSPENDED	STANDARD	Medium	0	0	N	Y	120
3	FINANCE_WH	SUSPENDED	STANDARD	Small	0	0	N	N	60
4	MARKETING_WH	SUSPENDED	STANDARD	X-Small	0	0	N	N	60
5	MY_WH	SUSPENDED	STANDARD	X-Small	0	0	N	N	60
6	SYSTEM$STREAMLIT_	SUSPENDED	STANDARD	X-Small	0	0	N	N	60

Part 2: Role Setup

Create three roles:

1. **MARKETING_ANALYST**
2. **FINANCE_USER**
3. **ETL_DEVELOPER**

Grant each role access only to its assigned warehouse.

ANS:

Creating MARKETING_ANALYST role

```
CREATE OR REPLACE ROLE MARKETING_ANALYST
    COMMENT = 'Role for Marketing Analysts';
```

Creating FINANCE_USER role

```
CREATE OR REPLACE ROLE FINANCE_USER
    COMMENT = 'Role for Finance Team';
```

Creating ETL_DEVELOPER role

```
CREATE OR REPLACE ROLE ETL_DEVELOPER
    COMMENT = 'Role for ETL Developers';
```

It was successfully executed.

Results (just now)

Table Chart Pivot

	status
1	Role MARKETING_ANALYST successfully created.

Results (just now)

Table Chart Pivot

	status
1	Role FINANCE_USER successfully created.

Results (just now)

Table Chart Pivot

	status
1	Role ETL_DEVELOPER successfully created.

	created_on	name	is_default	is_current	is_inherit	assigned_to_users	granted_to_roles
1	2026-05-12 21:59:39.723 -0700	ACCOUNTADMIN	Y	Y	N	1	0
2	2026-05-15 04:58:53.582 -0700	ETL_DEVELOPER	N	N	N	0	0
3	2026-05-15 04:58:30.230 -0700	FINANCE_USER	N	N	N	0	0
4	2026-05-15 04:52:44.888 -0700	MARKETING_ANALYST	N	N	N	0	0
5	2026-05-12 21:59:40.002 -0700	ORGADMIN	N	N	N	1	0
6	2026-05-12 21:59:39.732 -0700	PUBLIC	N	N	Y	0	0
7	2026-05-12 21:59:39.748 -0700	SECURITYADMIN	N	N	Y	0	1
8	2026-05-12 21:59:39.739 -0700	SYSADMIN	N	N	Y	0	1

Granting the usage of warehouses which are assigned to specified role

```
GRANT USAGE ON WAREHOUSE MARKETING_WH TO ROLE MARKETING_ANALYST;
GRANT USAGE ON WAREHOUSE FINANCE_WH   TO ROLE FINANCE_USER;
GRANT USAGE ON WAREHOUSE ETL_WH       TO ROLE ETL_DEVELOPER;
```

It was successfully executed.

Results (just now)

Table | Chart | Pivot

	status
1	Statement executed successfully.

Part 3: User Setup

Create users:

1. **MKT_USER1**
2. **FIN_USER1**
3. **ETL_USER1**

Set default role and default warehouse for each user.

ANS:

Creating MKT_USER1 user

```
CREATE OR REPLACE USER MKT_USER1
    PASSWORD          = 'MarketingPass123!'
    DEFAULT_ROLE      = MARKETING_ANALYST
    DEFAULT_WAREHOUSE = MARKETING_WH
    COMMENT           = 'Marketing Team User';
```

Creating FIN_USER1 user

```
CREATE OR REPLACE USER FIN_USER1
    PASSWORD          = 'FinancePass123!'
    DEFAULT_ROLE      = FINANCE_USER
    DEFAULT_WAREHOUSE = FINANCE_WH
    COMMENT           = 'Finance Team User';
```

Creating ETL_USER1 user

```sql
CREATE OR REPLACE USER ETL_USER1
    PASSWORD          = 'ETLPass123!'
    DEFAULT_ROLE      = ETL_DEVELOPER
    DEFAULT_WAREHOUSE = ETL_WH
    COMMENT           = 'ETL Developer User';
```

It was successfully executed.

(just now)

Table | Chart | Pivot

	status
1	User MKT_USER1 successfully created.

Results (just now)

Table | Chart | Pivot

	status
1	User FIN_USER1 successfully created.

Results (just now)

Table | Chart | Pivot

	status
1	User ETL_USER1 successfully created.

Results (just now)

Table | Chart | Pivot 4 rows 35ms

	name	created_on	login_name	display_name	first_name	last_name	email
1	ETL_USER1	2026-05-15 05:17:18.089 -0700	ETL_USER1	ETL_USER1	null	null	null
2	FIN_USER1	2026-05-15 05:16:59.398 -0700	FIN_USER1	FIN_USER1	null	null	null
3	MKT_USER1	2026-05-15 05:16:34.929 -0700	MKT_USER1	MKT_USER1	null	null	null
4	MS7039	2026-05-12 21:59:39.676 -0700	MS7039	MS7039	MANYA	SHRIVASTAVA	12amanyashrivastava@

Granting the roles which are assigned to specified user

```
GRANT ROLE MARKETING_ANALYST TO USER MKT_USER1;
GRANT ROLE FINANCE_USER      TO USER FIN_USER1;
GRANT ROLE ETL_DEVELOPER     TO USER ETL_USER1;
```

It was successfully executed.

Results (just now)

Table Chart Pivot

	status
1	Statement executed successfully.

must_change_password	snowflake_lock	default_warehouse	default_namespace	default_role	
lse	false	ETL_WH	null	ETL_DEVELOPER	["AL
lse	false	FINANCE_WH	null	FINANCE_USER	["AL
lse	false	MARKETING_WH	null	MARKETING_ANALYST	["AL
lse	false	COMPUTE_WH	null	ACCOUNTADMIN	["AL

Results (just now)

Table Chart Pivot 9 rows

	name	is_default	is_current	is_inherited	assigned_to_users	granted_to_roles	granted_roles
1	ACCOUNTADMIN	Y	Y	N	1	0	2
2	ETL_DEVELOPER	N	N	N	1	0	0
3	FINANCE_USER	N	N	N	1	0	0
4	MARKETING_ANALYST	N	N	N	1	0	0
5	ORGADMIN	N	N	N	1	0	0
6	PUBLIC	N	N	Y	0	0	0
7	SECURITYADMIN	N	N	Y	0	1	1

Part 4: Create Performance Demo Database

Create a database called:

RETAIL_MARKETING_DB

Create a table called:

CAMPAIGN_SALES

using Snowflake sample tables.

ANS:

Creating RETAIL_MARKETING_DB user while using ETL_WH warehouse

```
USE WAREHOUSE ETL_WH;

CREATE OR REPLACE DATABASE RETAIL_MARKETING_DB;
```

Creating RETAIL_MARKETING_DB.CAMPAIGN schema

```
CREATE OR REPLACE SCHEMA   RETAIL_MARKETING_DB.CAMPAIGN;
```

It was successfully executed.

Results (just now)

Table Chart Pivot

	status
1	Database RETAIL_MARKETING_DB successfully created.

Results (just now)

Table Chart Pivot

	status
1	Schema CAMPAIGN successfully created.

Results (just now)

Table Chart Pivot 4 rows 102ms

	created_on	name	is_default	is_current	origin	owner	comment
1	2026-05-15 05:23:09.733 -0700	RETAIL_MARKETING_DB	N	Y		ACCOUNTADMIN	
2	2026-05-12 21:59:40.213 -0700	SNOWFLAKE	N	N	SNOWFLAKE.ACCOUN		
3	2026-05-12 21:59:44.327 -0700	SNOWFLAKE_SAMPLE_DATA	N	N	SFSALESSHARED.SFC_	ACCOUNTADMIN	Preloaded TPCH Data
4	2026-05-12 22:23:58.931 -0700	USER$MS7039	N	N			

Creating CAMPAIGN_SALES table

```sql
CREATE OR REPLACE TABLE RETAIL_MARKETING_DB.CAMPAIGN.CAMPAIGN_SALES AS
SELECT
    ss.ss_ticket_number                          AS TICKET_NUMBER,
    ss.ss_customer_sk                            AS CUSTOMER_ID,
    c.c_first_name || ' ' || c.c_last_name        AS CUSTOMER_NAME,
    d.d_date                                     AS SALE_DATE,
    d.d_year                                     AS SALE_YEAR,
    d.d_month_seq                                AS SALE_MONTH,
    i.i_item_desc                                AS ITEM_DESCRIPTION,
    i.i_category                                 AS ITEM_CATEGORY,
    ss.ss_quantity                               AS QUANTITY_SOLD,
    ss.ss_sales_price                            AS SALE_PRICE,
    ss.ss_net_paid                               AS NET_PAID,
    ss.ss_net_profit                             AS NET_PROFIT,
    ss.ss_coupon_amt                             AS DISCOUNT_APPLIED

FROM   SNOWFLAKE_SAMPLE_DATA.TPCDS_SF10TCL.STORE_SALES ss
JOIN   SNOWFLAKE_SAMPLE_DATA.TPCDS_SF10TCL.CUSTOMER    c
       ON ss.ss_customer_sk = c.c_customer_sk
JOIN   SNOWFLAKE_SAMPLE_DATA.TPCDS_SF10TCL.DATE_DIM     d
       ON ss.ss_sold_date_sk = d.d_date_sk
JOIN   SNOWFLAKE_SAMPLE_DATA.TPCDS_SF10TCL.ITEM         i
       ON ss.ss_item_sk = i.i_item_sk
LIMIT  5000000;
```

It was successfully executed.

Results (just now)

Table Chart Pivot

	status
1	Table CAMPAIGN_SALES successfully created.

Results (just now)

Table Chart Pivot 13 rows 53ms

	name	type	kind	null?	default	primary key	unique key	check	exp
1	TICKET_NUMBER	NUMBER(38,0)	COLUMN	Y	null	N	N	null	null
2	CUSTOMER_ID	NUMBER(38,0)	COLUMN	Y	null	N	N	null	null
3	CUSTOMER_NAME	VARCHAR(51)	COLUMN	Y	null	N	N	null	null
4	SALE_DATE	DATE	COLUMN	Y	null	N	N	null	null
5	SALE_YEAR	NUMBER(38,0)	COLUMN	Y	null	N	N	null	null
6	SALE_MONTH	NUMBER(38,0)	COLUMN	Y	null	N	N	null	null
7	ITEM_DESCRIPTION	VARCHAR(200)	COLUMN	Y	null	N	N	null	null
8	ITEM_CATEGORY	VARCHAR(50)	COLUMN	Y	null	N	N	null	null
9	QUANTITY_SOLD	NUMBER(38,0)	COLUMN	Y	null	N	N	null	null
10	SALE_PRICE	NUMBER(7,2)	COLUMN	Y	null	N	N	null	null
11	NET_PAID	NUMBER(7,2)	COLUMN	Y	null	N	N	null	null
12	NET_PROFIT	NUMBER(7,2)	COLUMN	Y	null	N	N	null	null
13	DISCOUNT_APPLIED	NUMBER(7,2)	COLUMN	Y	null	N	N	null	null

Part 5: Grant Permissions

Grant proper database, schema, and table permissions to each role.

Marketing and Finance can only read data.

ETL Developer can create and modify tables.

ANS:

Granting the usage of databases and schemas which are assigned to specified role

```
GRANT USAGE ON DATABASE RETAIL_MARKETING_DB TO ROLE MARKETING_ANALYST;
GRANT USAGE ON DATABASE RETAIL_MARKETING_DB TO ROLE FINANCE_USER;
GRANT USAGE ON DATABASE RETAIL_MARKETING_DB TO ROLE ETL_DEVELOPER;

GRANT USAGE ON SCHEMA RETAIL_MARKETING_DB.CAMPAIGN TO ROLE MARKETING_ANALYST;
GRANT USAGE ON SCHEMA RETAIL_MARKETING_DB.CAMPAIGN TO ROLE FINANCE_USER;
GRANT USAGE ON SCHEMA RETAIL_MARKETING_DB.CAMPAIGN TO ROLE ETL_DEVELOPER;
```

It was successfully executed.

Results (2 minutes ago)

Table | Chart | Pivot

	status
1	Statement executed successfully.

Grant permissions of SELECT to role MARKET_ANALYST

```
GRANT SELECT ON ALL TABLES IN SCHEMA RETAIL_MARKETING_DB.CAMPAIGN
    TO ROLE MARKETING_ANALYST;
```

Grant permissions of SELECT to role FINANCE_USER

```
GRANT SELECT ON ALL TABLES IN SCHEMA RETAIL_MARKETING_DB.CAMPAIGN
    TO ROLE FINANCE_USER;
```

Grant permissions of all privileges to role ETL_DEVELOPER

```
GRANT ALL PRIVILEGES ON ALL TABLES IN SCHEMA RETAIL_MARKETING_DB.CAMPAIGN
    TO ROLE ETL_DEVELOPER;

GRANT CREATE TABLE ON SCHEMA RETAIL_MARKETING_DB.CAMPAIGN
    TO ROLE ETL_DEVELOPER;
```

It was successfully executed.

Results (2 minutes ago)

Table | Chart | Pivot

	status
1	Statement executed successfully. 1 objects affected.

Part 6: Performance Testing

Run a sales analysis query using XSMALL warehouse.

Then scale up the warehouse to SMALL.

Compare performance.

Then scale it down again.

ANS:

Running a sale analysis after changing the warehouse

```
USE WAREHOUSE MARKETING_WH;

SELECT
    ITEM_CATEGORY,
    SALE_YEAR,
    SUM(NET_PAID)                          AS TOTAL_REVENUE,
    SUM(QUANTITY_SOLD)                     AS TOTAL_UNITS,
    AVG(SALE_PRICE)                        AS AVG_PRICE,
    COUNT(DISTINCT CUSTOMER_ID)           AS UNIQUE_CUSTOMERS,
    SUM(NET_PROFIT)                        AS TOTAL_PROFIT,
    SUM(DISCOUNT_APPLIED)                  AS TOTAL_DISCOUNTS

FROM   RETAIL_MARKETING_DB.CAMPAIGN.CAMPAIGN_SALES
WHERE  SALE_DATE BETWEEN '2000-01-01' AND '2003-12-31'
GROUP  BY ITEM_CATEGORY, SALE_YEAR
ORDER  BY TOTAL_REVENUE DESC;
```

Resize the warehouse MARKETING_WH into SMALL

```
ALTER WAREHOUSE MARKETING_WH
    SET WAREHOUSE_SIZE = 'SMALL';
```

Rerun the query while setting cached result as FALSE and then resizing as XSMALL

```sql
ALTER SESSION SET USE_CACHED_RESULT = FALSE;

SELECT
    ITEM_CATEGORY,
    SALE_YEAR,
    SUM(NET_PAID)                          AS TOTAL_REVENUE,
    SUM(QUANTITY_SOLD)                     AS TOTAL_UNITS,
    AVG(SALE_PRICE)                        AS AVG_PRICE,
    COUNT(DISTINCT CUSTOMER_ID)           AS UNIQUE_CUSTOMERS,
    SUM(NET_PROFIT)                        AS TOTAL_PROFIT,
    SUM(DISCOUNT_APPLIED)                  AS TOTAL_DISCOUNTS

FROM   RETAIL_MARKETING_DB.CAMPAIGN.CAMPAIGN_SALES
WHERE  SALE_DATE BETWEEN '2000-01-01' AND '2003-12-31'
GROUP  BY ITEM_CATEGORY, SALE_YEAR
ORDER  BY TOTAL_REVENUE DESC;

ALTER WAREHOUSE MARKETING_WH
    SET WAREHOUSE_SIZE = 'XSMALL';
```

It was successfully executed.

Results (just now)

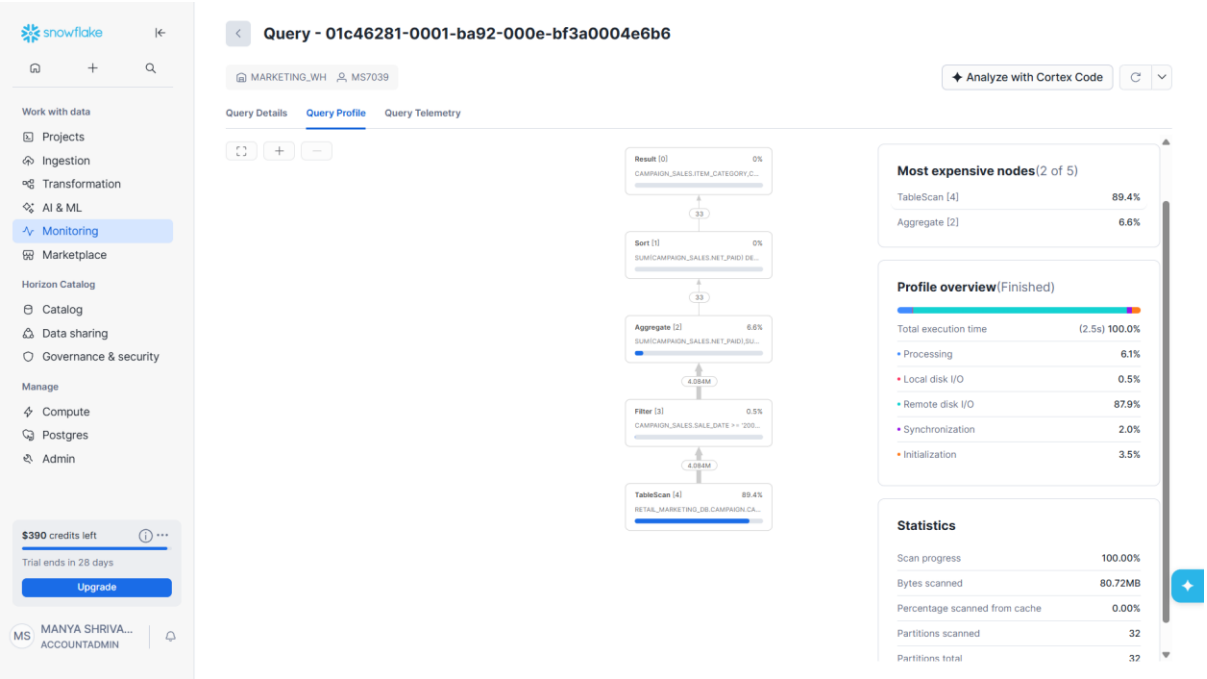
Table | Chart | Pivot

	status
1	Statement executed successfully.

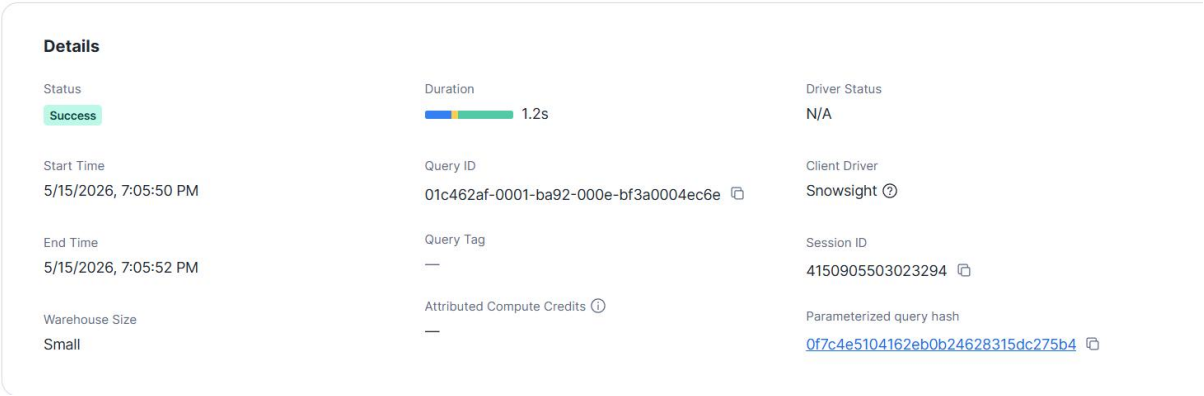
Query: 1 when warehouse size is XSMALL

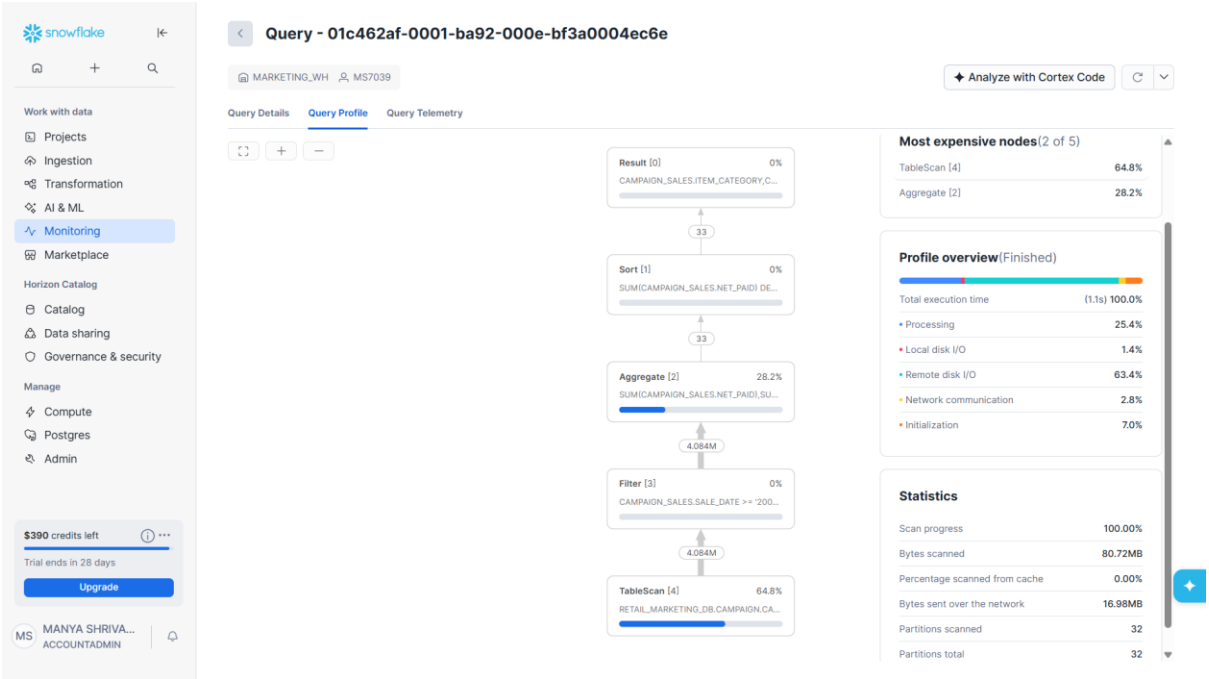
Details

Status: Success	Duration: 2.9s	Driver Status: N/A
Start Time: 5/15/2026, 6:19:13 PM	Query ID: 01c46281-0001-ba92-000e-bf3a0004e6b6	Client Driver: Snowsight
End Time: 5/15/2026, 6:19:16 PM	Query Tag: —	Session ID: 4150905503023138
Warehouse Size: X-Small	Attributed Compute Credits: —	Parameterized query hash: 0f7c4e5104162eb0b24628315dc275b4



Query: 2 when warehouse size is SMALL





Part 7: Cache Testing

Run the same query twice.

Observe why the second query is faster.

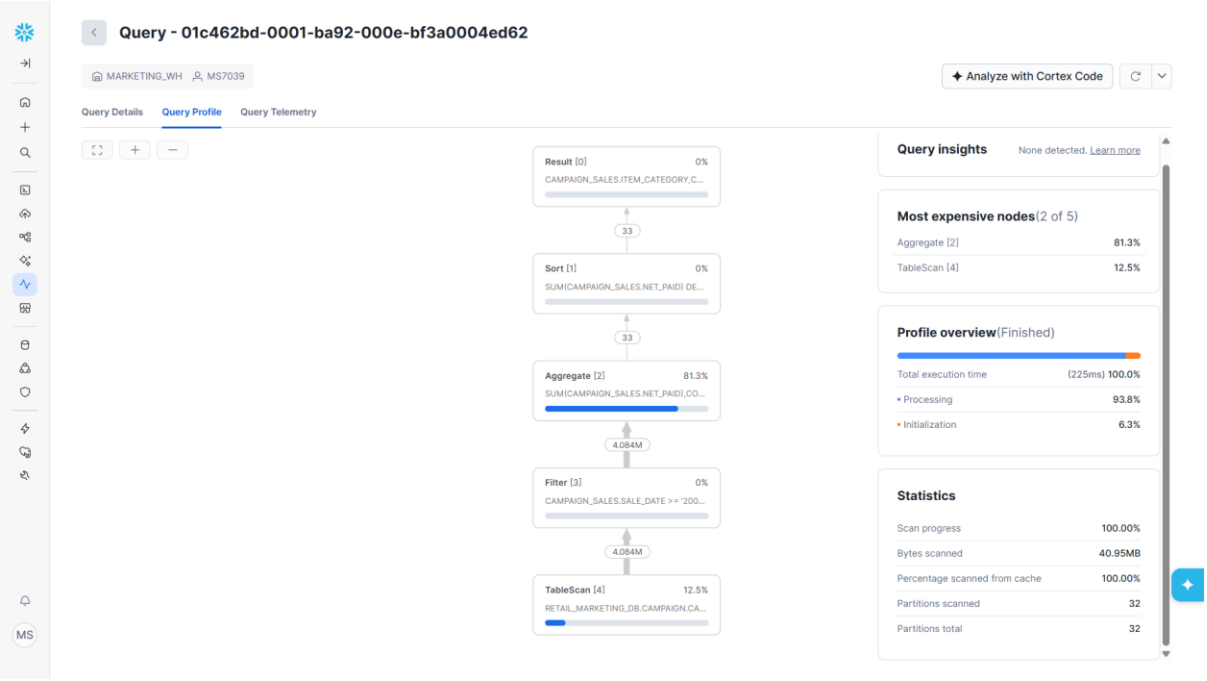
ANS:

The query is ran and rerun while setting cached result TRUE.

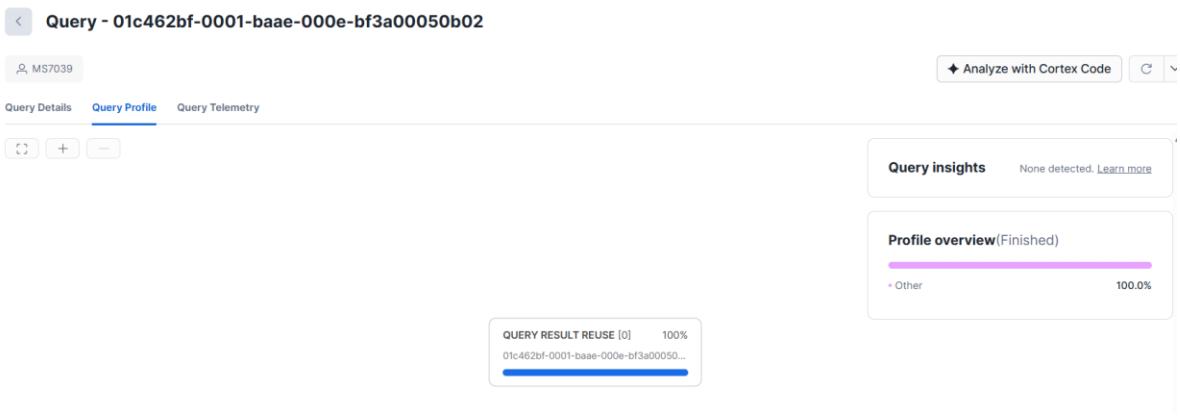
```sql
SELECT
    ITEM_CATEGORY,
    SALE_YEAR,
    SUM(NET_PAID)                    AS TOTAL_REVENUE,
    COUNT(DISTINCT CUSTOMER_ID)  AS UNIQUE_CUSTOMERS

FROM   RETAIL_MARKETING_DB.CAMPAIGN.CAMPAIGN_SALES
WHERE  SALE_DATE BETWEEN '2000-01-01' AND '2003-12-31'
GROUP  BY ITEM_CATEGORY, SALE_YEAR
ORDER  BY TOTAL_REVENUE DESC;
```

```sql
ALTER SESSION SET USE_CACHED_RESULT = TRUE;
| Ctrl+I to generate
SELECT
    ITEM_CATEGORY,
    SALE_YEAR,
    SUM(NET_PAID)                    AS TOTAL_REVENUE,
    COUNT(DISTINCT CUSTOMER_ID)  AS UNIQUE_CUSTOMERS

FROM   RETAIL_MARKETING_DB.CAMPAIGN.CAMPAIGN_SALES
WHERE  SALE_DATE BETWEEN '2000-01-01' AND '2003-12-31'
GROUP  BY ITEM_CATEGORY, SALE_YEAR
ORDER  BY TOTAL_REVENUE DESC;
```

It was successfully executed.



After rerun the result is taken from previous result at zero credit



Part 8: Cluster Key Testing

Run a date-filter query before clustering.

Add cluster key on SALE_DATE.

Run the same query again.

Compare performance.

ANS:

Base query without clustering while running date-time filter

```
SELECT
    SALE_DATE,
    ITEM_CATEGORY,
    SUM(NET_PAID)        AS DAILY_REVENUE,
    COUNT(*)            AS TRANSACTION_COUNT

FROM   RETAIL_MARKETING_DB.CAMPAIGN.CAMPAIGN_SALES
WHERE  SALE_DATE BETWEEN '2001-01-01' AND '2001-03-31'
GROUP  BY SALE_DATE, ITEM_CATEGORY
ORDER  BY SALE_DATE;
```

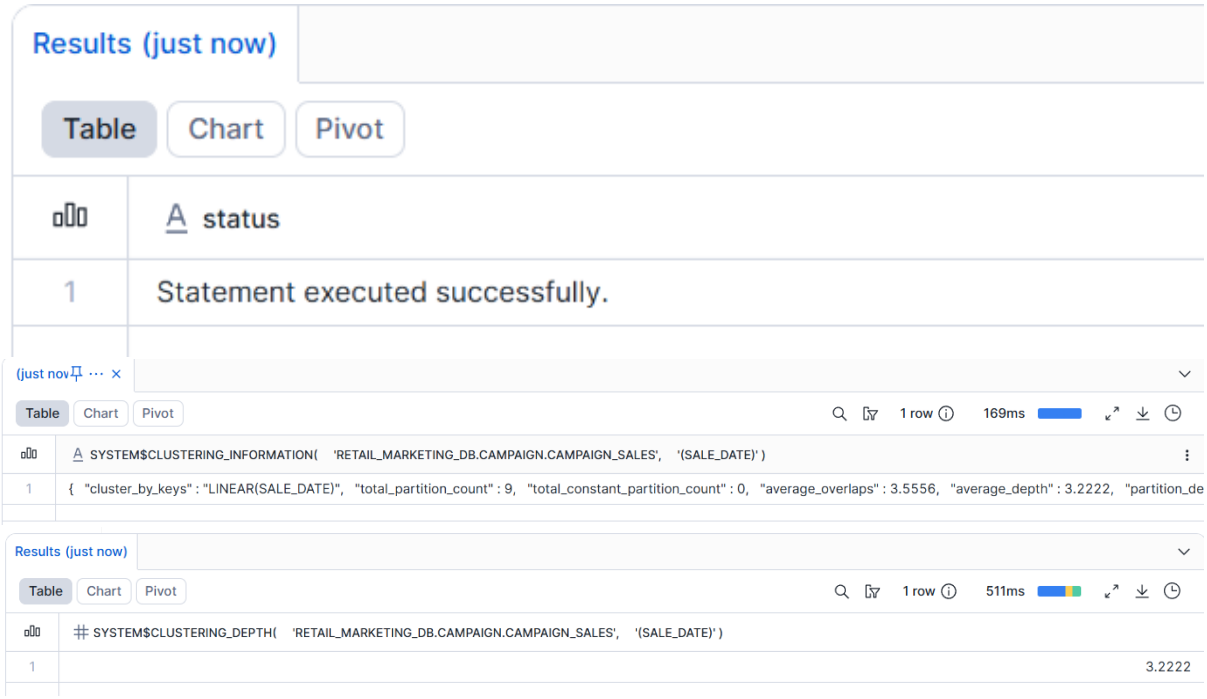
Adding cluster key on SALE_DATE

```sql
ALTER TABLE RETAIL_MARKETING_DB.CAMPAIGN.CAMPAIGN_SALES
    CLUSTER BY (SALE_DATE);

SELECT SYSTEM$CLUSTERING_INFORMATION(
    'RETAIL_MARKETING_DB.CAMPAIGN.CAMPAIGN_SALES',
    '(SALE_DATE)'
);

SELECT SYSTEM$CLUSTERING_DEPTH(
    'RETAIL_MARKETING_DB.CAMPAIGN.CAMPAIGN_SALES',
    '(SALE_DATE)'
);
```

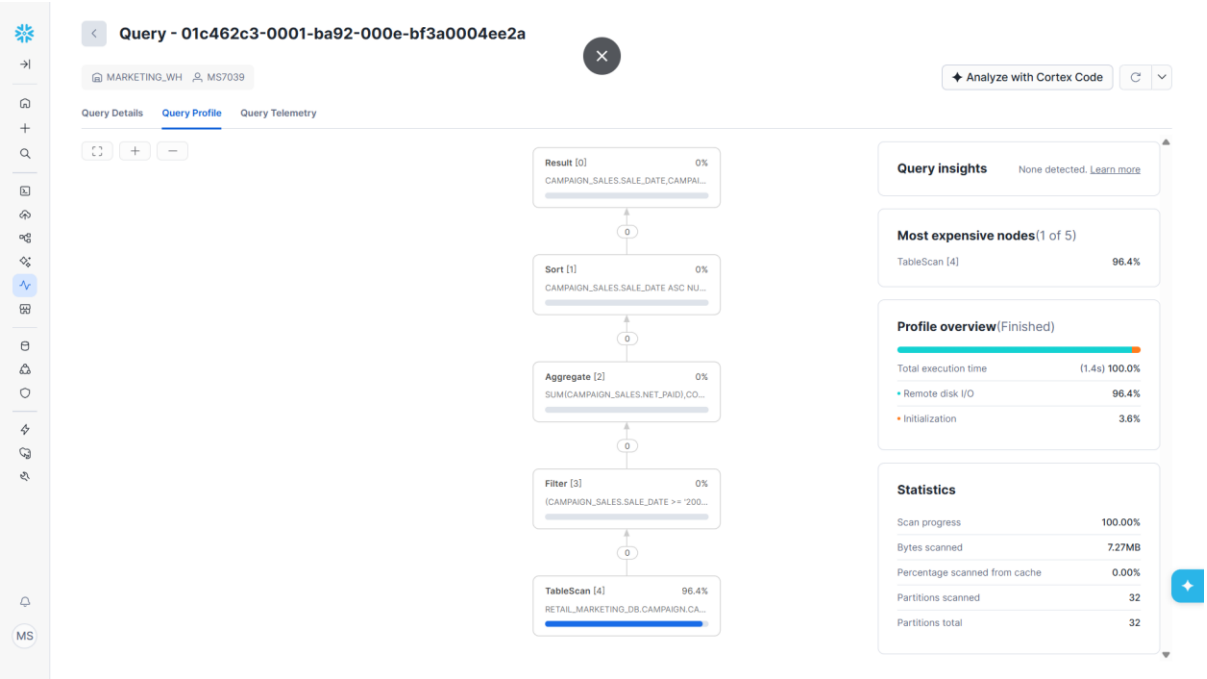
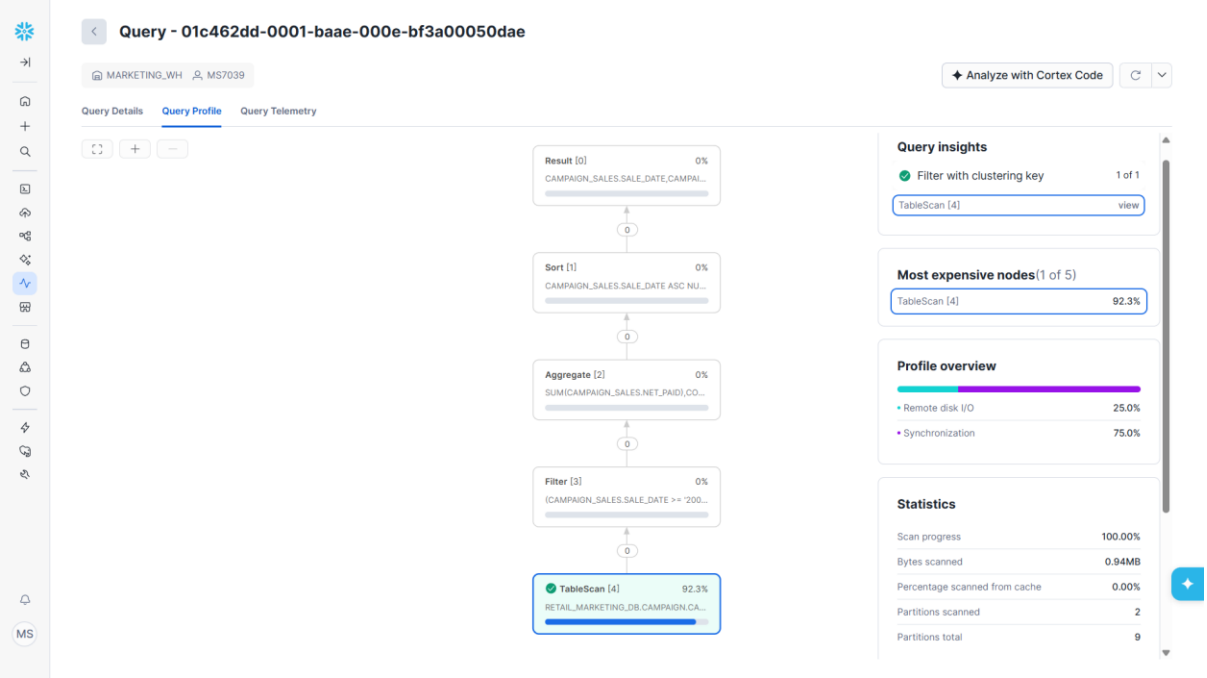
Base query with clustering while running date-time filter

```sql
SELECT
    SALE_DATE,
    ITEM_CATEGORY,
    SUM(NET_PAID)        AS DAILY_REVENUE,
    COUNT(*)            AS TRANSACTION_COUNT

FROM   RETAIL_MARKETING_DB.CAMPAIGN.CAMPAIGN_SALES
WHERE  SALE_DATE BETWEEN '2001-01-01' AND '2001-03-31'
GROUP  BY SALE_DATE, ITEM_CATEGORY
ORDER  BY SALE_DATE;
```

It was successfully executed.



Without clustering



With clustering